

- 7 **(a)** $V = 2.4 \text{ V} = V_p$
 $N_s / N_p = 50 = V_s / V_p$ M1
- $V_s = 2.4 \times 50 = 120\text{V}$
 $\text{Max } V_s = 120 \times \sqrt{2}$
 $= 170\text{V}$ A1
- (b)** $P_{\text{ave}} = \frac{1}{2} (\frac{1}{2}) P_0$
 $= \frac{1}{4} V_0^2 / R$
 $= \frac{1}{4} 170^2 / 47$ C1
 $= 154 \text{ W}$ A1
- (c)** $P_{\text{new}} = 2 P_{\text{ave}} = 307 \text{ W}$ (allow ecf part b) M1
- (d)** Direct voltage since the voltage shown is always positive (w.r.t. time) B1
- (e)** Transformers requires an input voltage that varies with time. B1

Section B

8. (a) (i) Area cut in time Δt is the curved surface area of a cylinder.